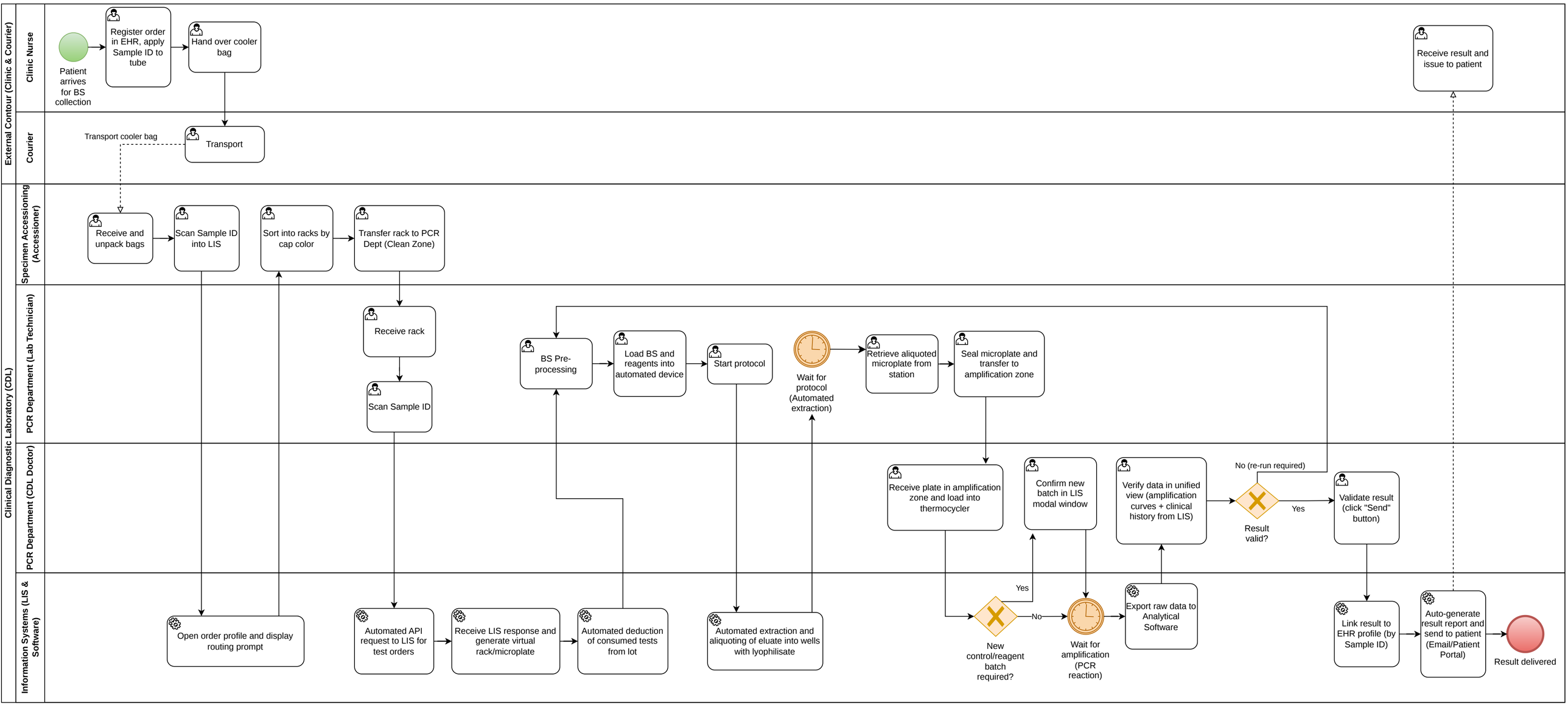
Business Process (To-Be Model): PCR Testing with Integrated LIS

Notation: BPMN 2.0 • **Participants:** Clinic, Courier, CDL (Accessioning, Lab Tech, CDL Doctor), Information Systems (LIS, Analytical Software) • **Version:** 2.0 (Portfolio)



Legend (BPMN 2.0)

- Start Event
- End Event
- User Task
- Service Task
- Timer Intermediate Event
- Exclusive Gateway (XOR)
- Sequence Flow
- Message Flow

Business Rules (To-Be Model)

- **Paperless workflow:** BS routing is executed exclusively via Sample ID barcode. Paper requisition forms and manual logbooks are completely eliminated.
- **Unified data environment:** Rack allocation and virtual microplate map generation are performed centrally in the LIS. Data is automatically synchronized with the EHR and Analytical Software.
- **Automated Lot Tracking:** Deduction of consumables (tests from a lot) is recorded automatically upon generation of the virtual microplate map.
- **Hardware integration:** LIS communicates directly with the robotic station and thermocycler via API (seamless two-way data exchange).

Process Risks and Mitigation

- **Integration failure (API down):** *Mitigation:* System must support fallback manual export/import of microplate maps (.csv/.xls formats) for loading into equipment via USB flash drive.
- **Unreadable Sample ID barcode on tube:** *Mitigation:* Accessioner and lab technician UI must include a field for manual keyboard entry of the Sample ID.
- **Reagent defect / invalid control results:** *Mitigation:* Strict software blocking of automated result delivery (CDL Doctor validation gateway) with mandatory routing of the process to re-aliquoting.